



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY
COLLEGE OF ENGINEERING

Department of Electro-Mechanical Engineering
Addis Machine and Spare Parts Manufacturing Industry

Internship Report

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Chapter 1

1. Brief History of Addis Machine and Spare Parts Manufacturing Industry (AMSMI)

1.1. Background of the company

The Addis Machine and Spare Parts Manufacturing Industry (AMSMI), previously known as Hibret Manufacturing and Machine Building Industry, has a rich history tied to Ethiopia's industrial and defense development. Founded in 1945 E.C. under Emperor Haile Selassie, it was established as part of a strategic effort to modernize Ethiopia's industrial capabilities.

The factory was created to support Ethiopia's vision of achieving military self-reliance and reducing dependency on foreign imports. Amid a shifting geopolitical environment, the need for a robust defense sector and industrial foundation became evident. Emperor Haile Selassie prioritized the development of domestic manufacturing to strengthen national sovereignty and sustain growth. AMSMI's initial focus was on producing small-scale military logistics, such as ammunition, medals, badges, and spare parts for military and industrial equipment, which were vital for bolstering Ethiopia's defense capabilities and national security.

1.1.1. International Collaboration

The factory was established through a partnership between the Ethiopian government and Czechoslovakia, capitalizing on the latter's expertise in industrial and defense manufacturing. A Czech national was appointed as the first manager, providing essential technical and managerial expertise for the factory's initial operations. This collaboration reflected Ethiopia's commitment to international partnerships to bolster its industrial sector, enabling critical technology and knowledge transfers that were foundational to the country's early manufacturing industry.

Initially, the factory focused on supplying Ethiopia's military with essential logistics, such as ammunition. Over time, as its potential became evident, the government expanded its role to support broader national industrialization goals.

1.1.2. Expansion and Diversification

As Ethiopia's industrial and military demands grew, AMSMI adapted accordingly. In the decades after its founding, the factory significantly expanded its manufacturing capabilities and economic role. By the 1970s and 1980s, it began producing civilian industrial machinery and spare parts alongside its military products, enabling AMSMI to contribute to both defense and civilian development.

In the early 2000s, Ethiopia pursued accelerated industrialization as part of its economic growth strategy, prompting reforms to modernize infrastructure and manufacturing. In 2002 E.C., the factory was renamed Addis Machine and Spare Parts Manufacturing Industry (AMSMI), reflecting its expanded mission to produce not only military equipment but also machinery and spare parts to support Ethiopia's burgeoning industrial sector.

1.1.3. Integration into METEC

To support Ethiopia's push for industrial development, AMSMI was incorporated into the Metals and Engineering Corporation (METEC) in 2002 E.C., a state-owned entity tasked with driving the nation's industrial transformation. METEC was formed as a major conglomerate to oversee Ethiopia's engineering and manufacturing sectors, aiming to advance industrialization by utilizing local resources, technology, and expertise to produce infrastructure, machinery, and equipment for both civilian and defense purposes.



Figure 1. 1 Logo of METEC

1.1.4. Challenges Faced Under METEC

Under METEC, AMSMI saw growth but faced significant challenges. By the late 2010s, METEC and its subsidiaries, including AMSMI, were criticized for mismanagement, corruption, and inefficiencies. METEC's role in major projects like the Grand Ethiopian Renaissance Dam (GERD) and sugar factories was marred by delays and financial irregularities, drawing public scrutiny.

In 2020, the Ethiopian government intervened, dissolving METEC as part of a broader reform to enhance transparency and accountability in public enterprises. This marked the end of METEC's tenure but provided AMSMI an opportunity to restructure and redefine its identity.

1.1.5. Rebranding as Part of Ethio Engineering Group

After METEC's dissolution in 2020, AMSMI was restructured and integrated into the newly established Ethio Engineering Group (EEG) to enhance efficiency and transparency in Ethiopia's industrial sector. Under EEG, AMSMI refocused on manufacturing industrial machinery and spare

parts for civilian and military use, emphasizing innovation, quality, and market expansion. By producing high-quality goods, AMSMI supports Ethiopia's industrialization goals, reduces reliance on imports, and strengthens local manufacturing, solidifying its role as a key player in the nation's economic and industrial growth.



Figure 1. 2: Ethio-Engineering Group logo

1.2.Main Products and Services:

AMSMI's Evolution as a Leading Producer of Industrial Machinery and Spare Parts

Addis Machine and Spare Parts Manufacturing Industry (AMSMI) has become a cornerstone of Ethiopia's industrial supply chain, emerging as a leading producer of industrial machinery and spare parts. Through innovation, expansion, and a commitment to quality, AMSMI supports diverse sectors, including manufacturing, agriculture, transport, and defense, contributing to Ethiopia's economic growth and industrial self-reliance. Its expertise in producing machinery and spare parts enhances operational efficiency across industries in Ethiopia and East Africa. Below are some of the primary products and services offered by AMSMI:

1.2.1. Industrial Machinery Production:

AMSMI manufactures a range of industrial machinery critical for sectors like manufacturing, construction, and agriculture. Its lathe machines enable precise cutting and shaping of metal for vehicle components, machinery, and infrastructure projects. Milling machines offer high precision for shaping materials used in machinery, construction, and agricultural equipment. Shearing machines provide efficient metal sheet cutting for metal fabrication and construction. Radial drill machines support heavy-duty drilling for large workpieces in engineering and manufacturing. Rolling machines shape metal plates for applications in automotive and construction industries.

1.2.2. Spare Parts Production:

AMSMI is a key producer of high-quality spare parts essential for machinery maintenance across various sectors, minimizing downtime and boosting productivity. Its gears, including spur, helical, and sprocket types, facilitate power transmission in agriculture, manufacturing, and automotive applications. Shafts deliver rotational force for vehicles and industrial machines. Molds and dies enable precise material shaping in plastics and metalworking industries. Brake pads ensure safe operation in vehicles and machinery, enhancing reliability and efficiency.

1.2.3. Fabrication Services:

AMSMI provides advanced fabrication services to meet diverse client needs, producing customized parts and structures for industries. Its steel structure fabrication supports Ethiopia's infrastructure development, creating durable components for buildings, bridges, and industrial facilities. Body structure fabrication delivers tailored solutions for industrial machinery and vehicles, ensuring efficient and safe operation. AMSMI also produces high-quality bolts and nuts, essential for assembling machinery, vehicles, and infrastructure projects, ensuring secure construction and maintenance.

1.2.4. Brake Pads and Related Products

As a leader in brake-related products, AMSMI ensures safety and efficiency in vehicles and industrial machinery. Its durable brake pads serve commercial vehicles, agricultural machinery,

and industrial equipment, offering reliable stopping power in demanding environments. AMSMI also produces brake lines and brake shoes, designed for heavy-duty use to maintain robust and safe braking systems.

1.2.5. Consultancy and Maintenance Services:

AMSMI provides consultancy and maintenance services to enhance machinery and production efficiency. Consultancy offers expert guidance to optimize performance. Maintenance services keep equipment in top condition, minimizing downtime and costs. Material testing ensures products meet high-quality standards. Heat treatment improves metal component strength and durability. Machine erection services handle proper installation and commissioning. Training programs equip technicians and engineers with skills to operate and maintain machinery effectively, reducing operational risks.

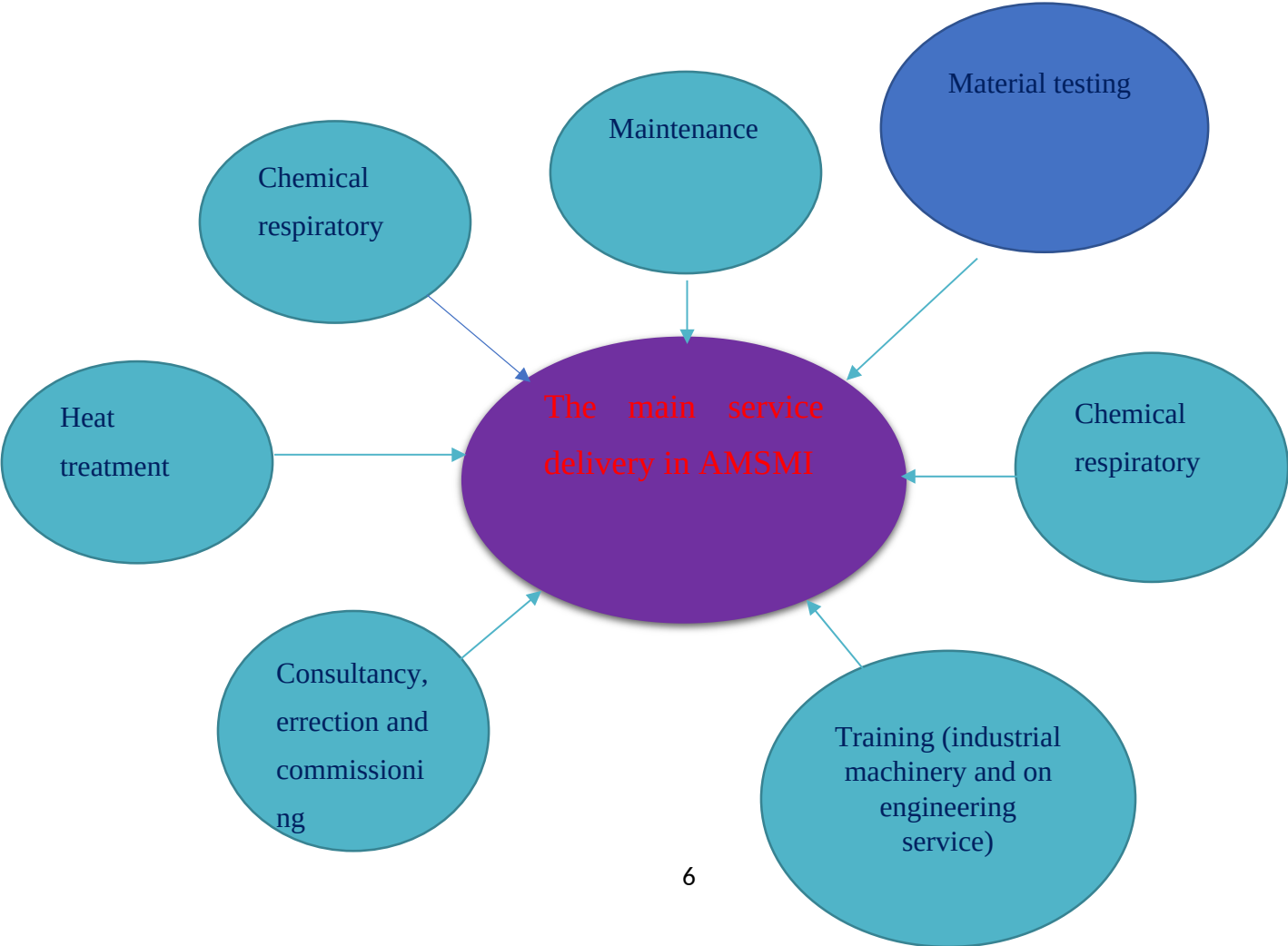


Figure 1. 3 : Visual Summary of Major Services of AMSMI

Industrial Machinery: AMSMI produces a wide range of industrial machinery to support Ethiopia's expanding industrial sector, including lathe machines for shaping metal, wood, and other materials; milling machines for precise cutting and shaping; shearing machines for cutting sheet metal; radial drill machines for heavy-duty drilling; and rolling machines for bending and forming metal plates and sheets.

Spare Parts: AMSMI specializes in manufacturing essential spare parts, such as gears (including worm, spur, helical, and sprocket types), shafts critical for machinery operation, molds, dies, and punches for industrial manufacturing, and brake pads vital for vehicles and industrial equipment.

Fabrication Services: AMSMI offers high-quality fabrication services, including steel structure fabrication for construction and industrial projects, body structure fabrication for industrial equipment, and production of bolts and nuts essential for machinery assembly.

Brake Pads: AMSMI is a leading manufacturer of brake pads, brake lines, and brake shoes for vehicles and industrial machinery, ensuring safety and efficiency in transportation and equipment operations.

Additional Services: AMSMI provides supplementary services, including consultancy for machinery setup and optimization, maintenance services for industrial equipment, material testing to ensure part quality and durability, heat treatment to enhance metal part strength, machine erection for installation and commissioning, and training programs to improve technicians' and engineers' skills in machinery operation and maintenance.



Figure 1. 4: Machinery Used to Perform the Above Listed Tasks

By offering such a wide range of products and services, AMSMI positions itself as a one-stop shop for the industrial and manufacturing needs of various sectors across Ethiopia.

1.3.Main Customers or End Users of Addis Machine and Spare Parts Manufacturing Industry (AMSMI)

AMSMI serves a broad and diverse customer base across multiple sectors, positioning itself as a critical partner for industries that depend on high-quality machinery and spare parts for their operations. Its products and services are essential to both the public and private sectors, contributing to the growth and sustainability of industries in infrastructure development, manufacturing, agriculture, and beyond. Below is a more detailed explanation of AMSMI's key customer segments and the roles they play in Ethiopia's industrial ecosystem:

1.3.1. Government Entities

AMSMI is a cornerstone of Ethiopia's public sector, supplying critical machinery and spare parts for government-led infrastructure, defense, and transportation projects. Its products, including lathe, milling, and drilling machines, as well as gears and brake pads, support major initiatives like road networks, dams, and railways, while also maintaining government transportation fleets, ensuring efficient public services.

In defense, AMSMI upholds its legacy by providing military logistics, such as spare parts for vehicles and machinery, bolstering national security. It also supports water management, energy, and agricultural projects, driving Ethiopia's modernization and infrastructure growth with essential equipment and components, cementing its role in the nation's industrial and economic progress.

1.3.2. Manufacturing Companies

AMSMI is a vital supplier to Ethiopia's growing manufacturing industry, providing high-quality spare parts, brake pads, and machinery to ensure consistent production and prevent costly downtime. Its reliable components and equipment enhance operational efficiency for manufacturers.

AMSMI delivers essential spare parts, including gears, shafts, bearings, and molds, to keep machinery operational, reducing financial losses from breakdowns. Its machinery, such as lathe machines, milling machines, and drill presses, supports precision manufacturing to meet domestic and export demands. AMSMI's brake pads, used in vehicles and industrial machinery, ensure safety and reliability. Additionally, its maintenance and consultancy services, including check-ups, repairs, and adjustments, prevent equipment failures, extend machinery lifespan, and maintain uninterrupted production schedules.

1.3.3. Agriculture and Agro-Processing

Ethiopia's agriculture sector, a vital component of the nation's economy, relies heavily on AMSMI for specialized machinery, equipment, and spare parts to boost productivity and efficiency in farming and agro-processing. AMSMI's contributions support modernization efforts, addressing the growing demand for food domestically and internationally.

AMSMI supplies durable farm machinery for plowing, harvesting, and irrigation, reducing labor and increasing yields for farmers. It provides essential spare parts, such as gears, hydraulic components, and engine parts, to maintain agricultural equipment under harsh conditions, minimizing downtime during critical seasons. For agro-processing, AMSMI delivers machinery and parts for processing, packaging, and storing agricultural products, supporting efficient production for domestic and export markets. Additionally, AMSMI offers cost-effective equipment for smallholder farmers, enhancing their livelihoods and aligning with Ethiopia's agricultural development goals.

1.3.4. International Markets

AMSMI is expanding its presence beyond Ethiopia, aiming to become a leading supplier of industrial machinery, spare parts, and brake pads in East Africa and beyond, supporting regional industrialization.

The company exports machinery and spare parts to countries like Kenya, Sudan, Djibouti, and South Sudan, aiding their manufacturing and infrastructure development while fostering regional economic growth. AMSMI's products support East Africa's infrastructure projects in construction, agriculture, and defense, establishing it as a key partner in the region's industrialization. Additionally, AMSMI is exploring new markets in North Africa, the Middle East, and Sub-Saharan Africa, leveraging its manufacturing expertise to meet diverse industrial needs and expand its global footprint.

1.4.Overall Organization and Workflow

AMSMI's organizational structure ensures efficient production and service delivery through specialized departments. The production department manufactures machinery, spare parts, and fabrication products, adhering to strict quality and delivery standards. The R&D department drives innovation, designing new products and enhancing existing ones to maintain market competitiveness. Marketing and sales manage customer relations, promote products, and explore new business opportunities. The finance department oversees budgeting, cost management, and financial stability, while HR handles recruitment, training, and employee development. The maintenance department minimizes downtime by keeping machinery operational. AMSMI's workflow follows a streamlined process: R&D designs products, production manufactures them to high standards, rigorous testing ensures quality, and products are delivered to clients, with ongoing improvements to machinery and skills to sustain a competitive edge.

Chapter 2

2. Overall internship experience

The internship at the machine building, machine body, and bolt and nut manufacturing factory provided us with an enlightening experience. We gained extensive insight into the operations of both bolt and nut production and machine body manufacturing. Despite lasting only three months, the internship offered us valuable hands-on experience. We were involved in nearly every aspect of the manufacturing process, which deepened our understanding of mechanical and electrical components. This exposure allowed us to appreciate the intricacies of production and the importance of precision in manufacturing. Additionally, we had learned about the various stages of production, from raw material selection to quality control measures. Observing the workflow and collaboration among different teams highlighted the significance of teamwork in achieving operational success. generally, this experience significantly enhanced our knowledge of the industry and the professional principles that guide a successful organization. The skills and insights we acquired will be invaluable as we move forward in our careers.

2.1.Sections of the Company we Worked In

Throughout our internship at Ethio Engineering Group, we had the opportunity to work in multiple sections of the company mostly in conventional and machine building factory's ranging from bolt and nut production to CNC machine maintenance and operation, gear and shaft production on the conventional store. We had been working in some the sections of the factory. The industrial unit had the following shops namely; bolt and nut manufacturing, machine body and structure building, machine building factory.

This exposure gave us a well-rounded experience in both the manufacturing and maintenance aspects of mechanical components. These hands-on experiences helped bridge the gap between our theoretical knowledge and practical application, significantly enhancing our skills.

2.1.1. Bolt and Nut Production section

In the bolt and nut production section, we was directly involved in the manufacturing process of these essential fasteners, which are crucial in vehicle and machinery assembly. My responsibilities included:

- Operating forging, threading, and cutting machines used to form bolts and nuts from raw materials.
- Ensuring quality control by regularly measuring and inspecting the produced components to meet precise specifications.
- Assisting with the maintenance of production machinery, ensuring they remained operational and efficient.

This experience helped us understand the critical role that precision manufacturing and quality assurance play in the mechanical components industry.

2.1.2. Gear and Shaft Production

In the gear and shaft production section, I worked on the manufacturing of gears and shafts—key components used in vehicles, machinery, and various industrial applications. This section allowed me to develop skills in:

- **Gear Cutting and Shaping:** I learned how to operate specialized machines such as gear hobbing machines, which cut teeth into the gears. This process requires high precision to ensure that the gears mesh properly with other mechanical parts.
- **Shaft Manufacturing:** I worked on lathes and grinding machines to shape and finish shafts, ensuring they met the required tolerances for smooth operation within gearboxes and other mechanical systems.
- **Heat Treatment Processes:** I gained knowledge about heat treatment, a crucial step in strengthening gears and shafts to enhance their durability and performance. This involved hardening the surface of components to resist wear and tear.

By working in this section, I gained a deep understanding of the critical role of gears and shafts in transmitting mechanical power. This experience also emphasized the importance of dimensional accuracy and surface finishes to ensure the longevity and performance of these components.

2.1.3. CNC Machine Maintenance

In the CNC machine maintenance section, I assisted with the repair and maintenance of advanced CNC machines used in cutting and shaping metal parts. My responsibilities included:

- **Performing routine maintenance** such as lubrication and calibration to keep the machines running smoothly.
- **Diagnosing and fixing mechanical and electrical issues**, including tool misalignment, motor faults, and wiring problems.
- **Assisting in programming** CNC machines to meet specific production requirements, improving my understanding of machine operation and automation.

This section honed my skills in troubleshooting and maintaining complex automated systems, giving me a strong foundation in both the mechanical and electrical aspects of CNC machines.

2.1.4. Machine building factory

In the machine building factory section, we worked on various tasks assigned by our instructor. One of the main tasks involved exploring motor control methods. These tasks allowed us to apply our theoretical knowledge in practical settings. We gained hands-on experience while learning about different techniques. The guidance from our instructor helped us understand the complexities of the tasks. Overall, this experience was valuable for our learning and skill development

- **Direct online motor control method**

The Direct On-Line (DOL) method is a straightforward way to start an electric motor by connecting it directly to the power supply. This method is suitable for small to medium-sized motors that require high starting torque.

Components Required: Motor, DOL starter (contactor and overload relay), Circuit breaker, Power supply, Control switch (start/stop)

Circuit Diagram:

We had drawn a circuit diagram showing connections between the power supply, DOL starter, motor, and control switch.

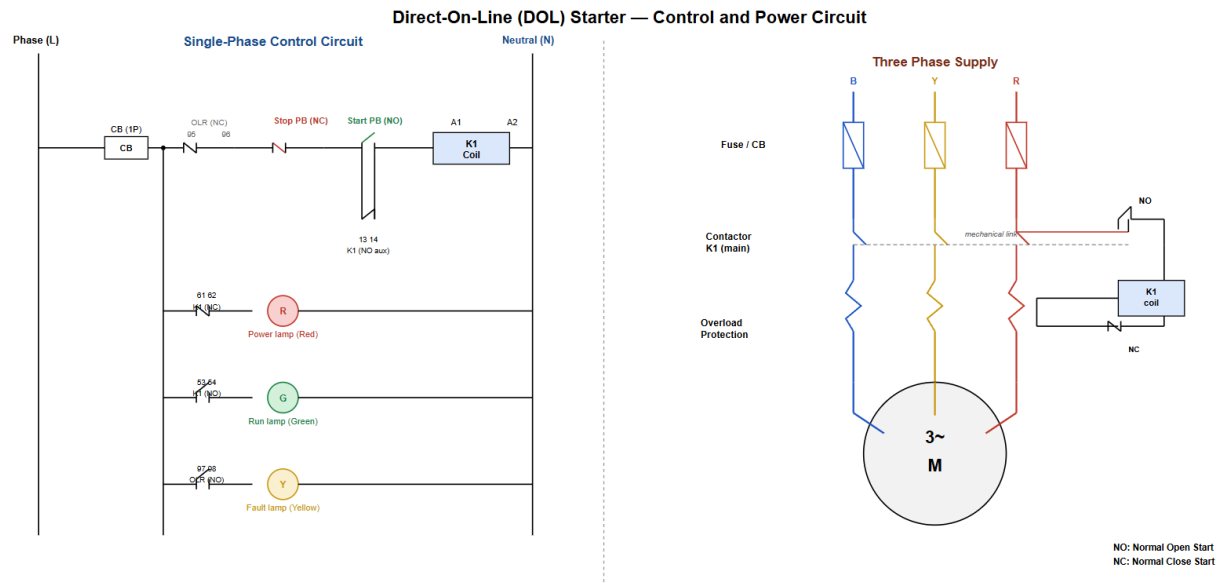


Figure 2. 1 Direct online motor control method Circuit Diagram

Procedures:

- I. We ensure all safety protocols are followed. Wear appropriate personal protective equipment (PPE).
- II. Wiring the circuits down then Connecting the three-phase power supply to the input terminals of the DOL starter.
- III. Connect the output terminals of the DOL starter to the motor terminals.
- IV. Install the overload relay in series with the motor to protect against overload conditions.
- V. We Connect a start and stop switch to the control circuit of the contactor. The start switch should be normally open (NO), and the stop switch should be normally closed (NC).
- VI. Double-check all connections for correctness and security, Ensure that the circuit breaker is in the "off" position before powering the system.
- VII. Press the start button to energize the contactor. This closes the circuit, allowing current to flow directly to the motor.

The motor should start immediately, reaching full speed quickly due to the direct connection.

- VIII. To stop the motor, we press the stop button. This opens the circuit and de-energizes the contactor, cutting power to the motor.

- **Forward and Reverse Motor Starting Method with Interlock Auxiliary Contacts**

The forward and reverse motor starting method with interlock auxiliary contacts ensures that the motor can run in both directions safely without the risk of both contactors being energized simultaneously. This is crucial to prevent motor damage.

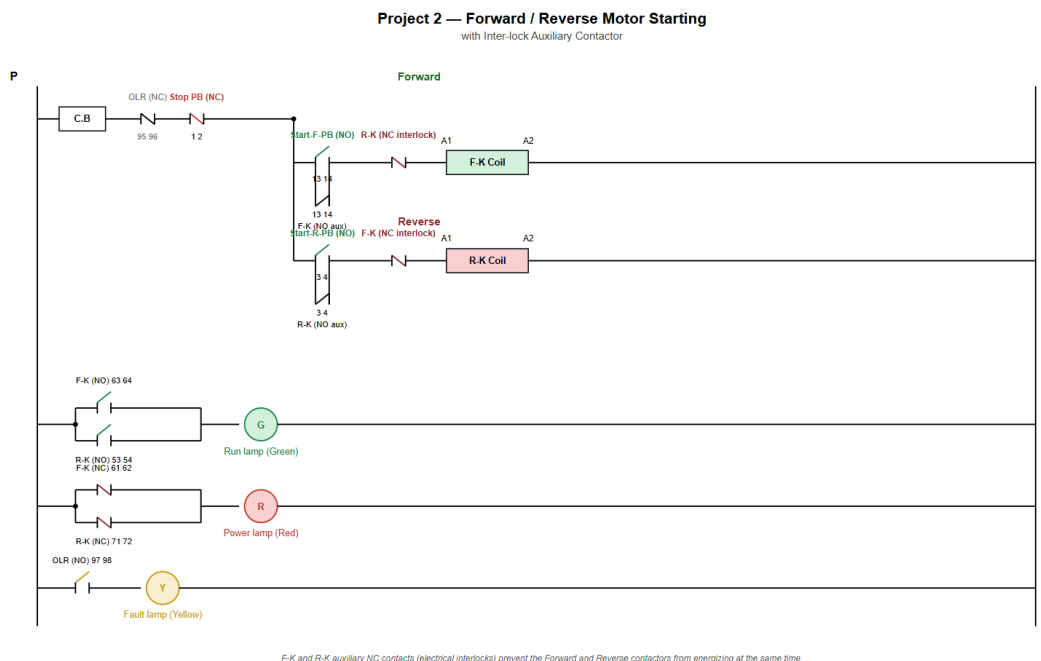


Figure 2. 2 Circuit Diagram of Forward and Reverse Motor Starting Method with Interlock Auxiliary Contacts

- **Interlocking with Auxiliary Contacts**

Auxiliary Contacts:

The forward contactor (F) has a normally open (NO) auxiliary contact that closes when the contactor is energized. The reverse contactor (R) has a normally open (NO) auxiliary contact that closes when it is energized.

Interlocking Configuration:

Connect the auxiliary contact of the forward contactor in series with the control circuit of the reverse contactor. Connect the auxiliary contact of the reverse contactor in series with the control circuit of the forward contactor. This configuration prevents both contactors from being energized simultaneously.

Procedure:

- I. Wiring the Circuit to connect the three-phase power supply to the circuit breaker. Then connect the output of the circuit breaker to both the forward and reverse contactors. Each contactor to the motor terminals through separate overload relays.
- II. Press the forward start button. The forward contactor energizes, closing its auxiliary contact, allowing current to flow to the motor, which starts rotating forward. The closed auxiliary contact of the forward contactor keeps the reverse contactor circuit open, preventing it from energizing.
- III. Press the reverse start button. The reverse contactor energizes, closing its auxiliary contact, allowing current to flow in the opposite direction, causing the motor to rotate in reverse. The closed auxiliary contact of the reverse contactor keeps the forward contactor circuit open.
- IV. To stop the motor, press the stop button. This opens the circuit for both contactors, cutting power to the motor.

2.1.5. Cross-Sectional Learning

Working across these diverse sections bolt and nut production, gear and shaft production, CNC machine maintenance, and various motor controlling methods gave me a comprehensive understanding of the company's operations. Each section provided valuable insights and practical skills that contributed to my development as a mechanical maintenance intern. These experiences helped me turn theoretical knowledge into practical expertise, allowing me to better understand how different mechanical systems work together to ensure the efficiency and quality of production processes.

2.2. Workflow in the Sections we worked in

During the internship at Ethio Engineering Group, we gained a firsthand understanding of the workflow in several sections, including bolt and nut production, gear and shaft production, and CNC machine maintenance. Each section followed a structured and coordinated process to ensure the smooth operation of manufacturing and maintenance activities.

2.2.1. Workflow in the Bolt and Nut Production Section

The bolt and nut production section followed a systematic manufacturing workflow to ensure the efficient production of high-quality fasteners:

- **Raw Material Preparation:** The workflow begins with raw materials, typically steel rods or bars, being delivered to the production area. These materials are checked for quality and specifications to ensure they meet the required standards.
- **Forging and Cutting:** The next step involves shaping the raw materials into bolts and nuts. Machines such as forging presses and **cutting machines** are used to create the basic shape of the components.
- **Threading:** After shaping, bolts are sent to threading machines, where precision threads are cut into the bolts. This process requires careful calibration to ensure the threads meet the correct size and pitch.
- **Quality Control:** Once the bolts and nuts are manufactured, they go through inspection to check for dimensional accuracy, surface finish, and threading precision. Any defective items are rejected or reworked.

- **Packaging and Distribution:** Finally, the finished products are packaged and sent to storage or directly to customers, depending on production schedules and orders.

Throughout the workflow, machine operators and quality control teams work closely to ensure that each step is completed efficiently and that the final products meet industry standards.

2.2.2. Workflow In the Machine Body and Structure and Manufacturing Section

- **Gear and Shaft Production section:** In the gear and shaft production section, the workflow was focused on precision manufacturing and involved several stages.
- **Material Selection and Preparation:** Similar to bolt and nut production, the process starts with selecting high-quality metal rods or bars, which are checked for specifications like hardness and tensile strength.
- **Gear Cutting and Shaping:** For gear production, the materials are sent to gear hobbing machines or gear shaping machines. These machines cut the teeth of the gears according to the design specifications. This process is highly precise, as even small deviations can affect the performance of the gears.
- **Shaft Turning and Finishing:** Shafts are manufactured using lathes and grinding machines. This process involves turning the raw material into the required shaft diameter and finishing the surface for smooth operation. Tolerances are tight, and precision is key to ensuring shafts are properly aligned within gearboxes.
- **Heat Treatment:** After the gears and shafts are shaped, they often undergo heat treatment to harden the surface and increase their durability. The components are heated in a furnace and then cooled in a controlled manner to enhance their mechanical properties.
- **Quality Control and Inspection:** As with other production processes, gears and shafts go through a strict quality control process. Dimensional accuracy, surface finish, and hardness are checked to ensure the parts meet the required standards before they are approved for assembly or delivery.
- **Assembly or Shipment:** Once the parts pass inspection, they are either sent for assembly into larger systems (such as gearboxes) or packaged and shipped to customers or other sections within the company.

2.2.3. Workflow in the CNC Machine Maintenance Section

The workflow in the CNC machine maintenance section was more dynamic, focusing on regular maintenance and urgent repairs to keep the machines running smoothly.

- **Scheduled Preventive Maintenance:** The maintenance team follows a preventive maintenance schedule for CNC machines. This involves routine checks, lubrication of moving parts, calibration of sensors, and cleaning to prevent malfunctions. Preventive maintenance is critical for avoiding costly breakdowns.
- **Diagnosis and Troubleshooting:** When a machine malfunctions, the first step is to diagnose the problem. This involves testing the electrical circuits, inspecting mechanical components, and running diagnostic software. Technicians use tools like multimeters and calipers to identify electrical or mechanical issues.
- **Repair and Replacement:** Once the issue is identified, the team proceeds with the repair. This could involve replacing worn-out parts, adjusting the alignment of mechanical components, or recalibrating the machine's controls. For electrical issues, technicians might rewire circuits or replace faulty sensors or motors.
- **Testing and Calibration:** After repairs are completed, the machine undergoes a series of tests to ensure its functioning correctly. This involves running the machine in a controlled environment and checking the output for precision. If necessary, further adjustments are made to calibrate the machine for optimal performance.
- **Documentation and Reporting:** Once the maintenance is complete, the work is documented, including the details of the repairs and any parts that were replaced. This ensures that the machine's maintenance history is recorded for future reference.

Across all sections, there was a strong emphasis on collaboration between teams to ensure efficient operations. For example:

- The quality control team worked closely with machine operators to ensure that products met specifications.
- The maintenance team collaborated with production teams to minimize downtime by performing maintenance during off-peak hours or when machines were not in use.
- Regular meetings and communication between departments ensured that production targets were met and that maintenance schedules did not interfere with operations.

This structured workflow, combined with teamwork and communication, ensured that each section of the company operated smoothly and efficiently.

2.3. Procedures and Methods Used During the Internship

During our internship we followed specific procedures and methods to ensure that the tasks we performed were done safely, efficiently, and correctly. These procedures applied to both mechanical and electrical aspects of the electro-mechanical systems we worked on.

2.3.1. Diagnostic and Troubleshooting Methods

For both electrical and mechanical issues, I followed a structured troubleshooting approach:

- **Visual Inspection:** we always began by visually inspecting the machinery or electrical systems to identify any obvious signs of wear, damage, or loose connections. This included checking for loose wires, burnt components, or worn-out mechanical parts.
- **Testing and Measurement Tools:** For electrical work, we used tools like multimeters to measure voltage, current, and resistance across electrical circuits. This helped me determine if a component had failed or if there were issues such as short circuits or power fluctuations. For mechanical systems, we used tools like torque wrenches to ensure components were properly tightened and maintained alignment.
- **Systematic Problem-Solving:** we approached problems systematically, starting with basic checks (e.g., ensuring power supply is connected or checking lubrication levels) before moving on to more complex diagnostics like tracing electrical faults or disassembling mechanical components.

2.3.2. Maintenance Procedures

- **Preventive Maintenance:** we followed regular preventive maintenance schedules to keep machinery in good working condition. This included tasks such as:

- Lubricating moving parts to reduce friction and wear.
 - Checking electrical wiring for loose connections or signs of overheating.
 - Inspecting belts, gears, and bearings for wear and replacing them if necessary.
- **Disassembly and Reassembly:** For mechanical maintenance tasks, we followed step-by-step procedures to disassemble parts of machinery, clean or repair them, and reassemble them. we carefully followed the manufacturer's instructions (or translated manuals when necessary) to ensure each component was correctly aligned and reassembled without causing damage to the system.

2.3.3. Electrical Work Procedures

For tasks involving electrical components, we adhered to the following key safety and operational procedures:

- **De-energizing Systems:** Before performing any electrical repairs or maintenance, we always ensured that the system was properly shut down and disconnected from the power source to avoid electric shock or damage to the equipment.
- **Use of Proper Safety Gear:** we used appropriate personal protective equipment (PPE), such as insulated gloves and safety glasses, when working on live electrical systems to minimize the risk of injury.
- **Wiring and Circuit Testing:** When repairing or replacing electrical components, we followed proper wiring diagrams to ensure the correct connections were made. After repairs, we used testing equipment like continuity testers to verify that circuits were complete and safe to use.

2.3.4. Safety Compliance and Standard Operating Procedures (SOPs)

- **Following Manufacturer's Guidelines:** we followed the manufacturer's standard operating procedures (SOPs) when working on machinery or electrical systems. These guidelines were crucial for ensuring that maintenance and repair tasks were performed safely and in accordance with the specifications provided by the equipment's manufacturer.
- **Adhering to Workplace Safety Protocols:** Safety was a priority, we complied with workplace safety protocols, such as using proper lifting techniques when handling heavy equipment, locking out/tagging out electrical systems before starting work, and wearing appropriate PPE like helmets, gloves, and safety boots.

These procedures ensured that the tasks we performed during my internship were carried out efficiently, safely, and according to industry standards.

2.4. Challenges Faced

2.4.1. Lack of Practical Knowledge

As a student entering the workforce for the first time, one of the primary challenges we faced was the gap between theoretical knowledge and practical application. While we had studied both electrical and mechanical systems in the classroom, applying this knowledge in real-world scenarios was difficult.

For example, when we were asked to troubleshoot electrical issues in machinery, we found that the complexity of the systems was far beyond what we had encountered during our studies. The integration of electrical circuits with mechanical components meant that a fault could arise from either system, and diagnosing the root cause required both mechanical and electrical expertise. Initially, we lacked the hands-on skills to perform such tasks efficiently.

We often needed guidance from my supervisors, who demonstrated the proper techniques for identifying and fixing issues. Over time, by observing and assisting senior technicians in both electrical and mechanical tasks such as rewiring electrical panels or performing routine maintenance on motors became more confident. However, this experience highlighted the importance of practical training in preparing for work in an industrial setting.

2.4.2. Limited or No Experience with Machinery

We were introduced to various types of machinery, such as lathe machines, milling machines, heavy-duty motors, and many of which were entirely new for us. These machines were often complex, requiring both mechanical adjustments and electrical wiring repairs. For instance, maintaining motor-driven systems involved not only mechanical alignment but also checking the electrical circuits to ensure proper voltage flow and signal control.

Given my lack of experience, even basic tasks, such as disassembling parts for maintenance or troubleshooting electrical faults, were initially challenging. We were unfamiliar with the operational procedures and safety protocols for handling such equipment. Additionally, the machinery was often large and intricate, which made it difficult to identify which components were causing the problem.

To overcome this, we took a proactive approach by requesting additional training and shadowing senior engineers as they worked on more complex machines. We also dedicated extra time outside working hours to study manuals and read about the machinery we was working with. Over time, this hands-on experience helped me gain a better understanding of both the mechanical and electrical components of the systems, allowing us to perform my duties more effectively.

2.4.3 Communication and Collaboration Challenges

Communication was another significant challenge during my internship. The engineering team often used technical terms and industry jargon related to both electrical and mechanical processes, which we struggled to fully understand during meetings and hands-on training sessions. Discussions on topics like electrical wiring diagrams, circuit breakers, and motor performance specifications were essential for troubleshooting electromechanical issues, but we found ourselves getting lost in the technical language.

For example, during one troubleshooting session for a faulty conveyor system, we had difficulty following the explanation of how to trace electrical wiring and test the motor's mechanical components. This made collaboration with more experienced team members challenging, as we occasionally misunderstood their instructions or the technical process they were describing.

To resolve this, we made an effort to ask questions whenever we encountered unfamiliar terminology and took notes during technical meetings. We also conducted personal research on electrical and mechanical terms, which helped us become more familiar with the industry language. Over time, this proactive approach improved my communication skills and enabled us to collaborate more effectively with the team.

2.4.4. Physical Demands and Safety Compliance

The electromechanical work we was involved in often required physically demanding tasks, including lifting heavy components, working in tight or awkward spaces, and handling complex tools. For example, rewiring electrical systems or adjusting mechanical parts often required working in confined areas or climbing onto machines, which presented both physical and technical challenges. Additionally, handling tools such as torque wrenches or multimeters required not just physical strength but also a good understanding of how to use them safely and efficiently.

Initially, we found it difficult to manage the physical strain and handle heavy equipment properly. The work was not only physically taxing but also required careful attention to safety procedures to avoid accidents, especially when dealing with live electrical systems.

To address this, we observed how the senior technicians handled these tasks and followed their guidance on using proper lifting techniques and handling tools safely. Over time, we developed better physical endurance and learned how to use the correct body mechanics when lifting or working with heavy machinery, significantly improving my efficiency and safety on the job.

2.4.5. Difficulty with Machinery Manuals in Chinese

A significant challenge we encountered was that many of the machinery manuals were written in Chinese, particularly for equipment sourced from overseas manufacturers. These manuals contained important instructions on both the electrical and mechanical components of the machines, including wiring diagrams, operational guidelines, and maintenance procedures. The language barrier made it extremely difficult to understand these instructions, which led to delays in troubleshooting and maintenance tasks.

For instance, when servicing an imported motor assembly system, we struggled to follow the operational procedures because the manual was in Chinese. This increased the risk of errors, as we could not fully understand the safety warnings or the proper sequence of steps to follow.

To overcome this, we resorted to using online translation tools, which allowed me to roughly translate key sections of the manuals. However, this process was time-consuming, and the translations were not always accurate, which further decreased my efficiency. Despite these difficulties, we persisted by combining the translations with advice from more experienced technicians who were familiar with the equipment. Although this method slowed down my work, it allowed us to complete my tasks with fewer mistakes.

2.4.6. Strategies for Overcoming Challenges

To overcome the challenges, we faced during my internship, we employed a variety of strategies. First, we sought continuous guidance from my supervisors and senior colleagues, allowing me to bridge the gap between theory and practice. Observing their techniques helped us understand both the mechanical and electrical aspects of the machinery.

For communication challenges, we asked questions to clarify technical terms and conducted additional research outside of work. This helped us improve my understanding of technical language, allowing us to better participate in discussions and collaborate with my team.

To address the physical demands of the work, we focused on learning proper techniques for handling heavy tools and equipment, while also adhering to safety protocols. Observing professionals around us and following their best practices enabled me to handle the physical aspects of the job more confidently.

Lastly, to mitigate the language barrier in understanding machinery manuals written in Chinese, we used online translation tools, and although it decreased my efficiency, it allowed us to complete the tasks without major errors. In the future, we plan to explore more efficient translation methods or seek help from colleagues proficient in the language.

2.5. Conclusion

our internship involved working on complex electromechanical systems, which presented a variety of challenges, including a lack of practical knowledge, limited experience with machinery, communication difficulties, physical demands, and language barriers with Chinese manuals. However, by proactively seeking assistance, improving my technical knowledge, and adopting better physical and safety practices, we were able to overcome most of these challenges.

This experience helped us develop both electrical and mechanical skills, improved my ability to collaborate with others, and highlighted the importance of continuous learning. Moving forward, we feel better equipped to handle similar challenges in the future and contribute effectively to electromechanical maintenance and repair work.

Chapter Three

3. Overall Internship Achievements and Experience

During the course of the internship, various skills and knowledge were acquired through participation in significant projects, including the design and development of a movable oil pumping tank. The internship provided an essential platform for professional growth, with opportunities to apply theoretical knowledge in real-world industrial settings. The overall achievements and skills gained are detailed below:

3.1. Theoretical Knowledge Was Upgraded

The application of theoretical knowledge gained from academic studies was facilitated in real-world scenarios during the internship.

- **Engineering Principles Were Applied:** Key engineering principles such as mechanical design, fluid dynamics, and lubrication systems were applied, contributing to the success of the movable oil pumping tank project. Theories from thermodynamics and material science were utilized to select the proper materials and pump systems for the project.
- **Industrial Systems Were Understood:** Insights into industrial systems, including supply chain management and production workflows, were gained. The interconnectedness of design, production, and operational efficiency was better understood through practical exposure to these processes.

3.2. Practical Skills Were Improved

Practical skills were significantly enhanced, bridging the gap between theory and application throughout the internship.

- **Design and Drive Systems Were Learned:** Hands-on experience was gained in designing mechanical systems and drive mechanisms. Work was carried out on the design of the pump system and mobility features of the oil tank, involving the selection of appropriate motors and wheels for heavy-duty operations.

- **Soft-Computing Applications Were Used:** SOLIDWORKS software was utilized for creating and testing prototypes, while simulation software was explored for predicting the design's performance under operational conditions.
- **Exposure to Industry Trends:** Emerging trends such as Industry 4.0 and automation were introduced, providing exposure to how AI is being applied in predictive maintenance and other modern manufacturing practices

3.3. Industrial Problem-Solving Capability Was Enhanced

Real-world challenges within the company contributed to the development of industrial problem-solving skills.

- **Innovative Solutions Were Developed:** The movable oil pumping tank was designed to address inefficiencies in the manual oil refilling process, improving lubrication efficiency and safety in the workplace.
- **Critical Thinking Was Applied:** Problem-solving and troubleshooting skills were strengthened by identifying machinery issues and optimizing workflows, which led to reduced downtime and increased efficiency.

3.4. Team Working Skills Were Strengthened

Collaboration with different departments and team members during the internship contributed to improved teamwork skills.

- **Interdepartmental Collaboration Was Experienced:** Regular collaboration occurred with departments such as production, maintenance, and quality control to ensure seamless project execution.
- **Conflict Resolution Was Practiced:** Conflicting ideas were navigated, and compromises were reached, enhancing the understanding of teamwork and collaboration dynamics within the company.

3.5. Leadership Skills Were Gained

Opportunities to demonstrate leadership were presented during the internship, despite the intern role.

- **Project Management Was Practiced:** Leadership was exercised by organizing team meetings, delegating tasks, and ensuring deadlines were met during the movable oil pumping tank project.
- **Team Motivation Was Provided:** Team members were guided and supported throughout the project, emphasizing the importance of leadership in fostering collaboration and maintaining a motivated work environment.

3.6. Work Ethics Were Understood

The internship provided a solid foundation for understanding and practicing professional work ethics.

- **Professionalism Was Demonstrated:** Professional conduct was maintained, including punctuality, responsibility, and adherence to company policies and procedures. Safety protocols were followed rigorously when working with machinery.
- **Accountability Was Learned:** Responsibility for project outcomes was encouraged, and diligence in completing tasks was practiced to ensure successful results in the industrial setting.

3.7. Entrepreneurship Skills Were Acquired

Entrepreneurial thinking was fostered through exposure to business-related challenges during the internship.

- **Innovation Was Encouraged:** Problem-solving and innovation were viewed as key to creating value within the company. The movable oil pumping tank project exemplified how innovative solutions can lead to operational improvements and cost savings.
- **Resource Management Was Observed:** Resource management skills were developed through observation of how project materials, manpower, and time were optimized to achieve the best outcomes.

Chapter Four

4. Conclusion & Recommendation

4.1. Conclusion

The internship experience at the Addis Machine and Spare Parts Manufacturing Industry (AMSMI) proved to be an invaluable opportunity for immersive learning and professional growth. Through active participation in various sections of the factory such as bolt and nut production, gear and shaft manufacturing, and CNC machine maintenance we were able to gain hands-on experience that significantly bridged the gap between theoretical knowledge acquired in academic settings and practical applications in a real-world manufacturing environment. This direct involvement not only allowed us to operate sophisticated machinery but also provided insights into critical processes such as quality control and workflow management.

One of the most striking aspects of the internship was the emphasis on teamwork and collaboration. Working alongside skilled professionals highlighted the importance of effective communication and cooperation in achieving operational success. We learned that the manufacturing sector relies heavily on the collective efforts of diverse teams, each contributing unique skills and perspectives to the production process. This realization deepened our appreciation for the complexity of manufacturing operations and underscored the necessity of fostering a positive workplace culture.

Additionally, the internship offered us insights into the broader industrial landscape in Ethiopia, particularly the role of local manufacturing capabilities in supporting national development goals. As we observed the processes involved in producing machinery and spare parts essential for various sectors, it became evident that companies like AMMSI are vital to enhancing Ethiopia's industrial self-sufficiency. This experience reinforced our belief in the significance of investing in local industries to drive economic growth and reduce dependence on imports.

The challenges encountered during the internship, such as adapting to the fast-paced environment and mastering complex machinery, further contributed to our personal and professional development. These experiences taught us resilience and the importance of continuous learning skills that will undoubtedly serve us well in our future careers.

In summary, the internship at AMMSI was a transformative experience that equipped us with practical skills, deepened our understanding of the manufacturing process, and reinforced the importance of local industrial capabilities. The knowledge gained during this time will serve as a strong foundation as we embark on our professional journeys, instilling in us a commitment to contributing positively to Ethiopia's industrial landscape and beyond. As we move forward, we carry with us not only technical skills but also a profound appreciation for the collaborative spirit and innovative potential that define the manufacturing sector.

4.2. Recommendation:

As interns in the Addis Machine and Spare Parts Manufacturing Industry (AMSMI) we recommend the following steps for future interns who are seeking to make the most of their internship experience and also for the company to further enhance the efficiency of their product.

4.2.1 Recommendation for Future Internes:

For students seeking to make the most of their internship experience, we offer the following recommendations based on our own experience:

- Based on our collective experience during the internship at Addis Machine and Spare Parts Manufacturing Industry (AMSMI), we highly recommend this type of industrial placement for aspiring mechanical and industrial engineers. The wide variety of tasks, combined with opportunities to collaborate with seasoned professionals, provides an ideal setting for interns to gain practical, hands-on experience in the field of engineering and industrial manufacturing.
- During our time at AMSMI, we were consistently exposed to an extensive range of industrial processes that allowed us to apply theoretical knowledge in a practical setting. We were tasked with operating crucial machinery such as lathe and milling machines, and were also actively involved in machine maintenance activities. Additionally, we contributed to important design projects, such as the creation of a movable oil pumping tank.

These experiences deepened our understanding of the machinery and processes fundamental to industrial production. We believe that this type of exposure is essential for bridging the gap between academic studies and practical industrial applications. Future interns can expect

a highly practical and hands-on environment, where they will be encouraged to apply their classroom knowledge to real-world engineering challenges.

- One of the significant benefits of the AMSMI internship is the opportunity to develop advanced problem-solving skills. The tasks assigned to us often involved troubleshooting and optimizing mechanical systems, which helped enhance our critical thinking abilities. For example, we worked on the design and improvement of mechanical systems, which required us to identify issues, explore alternative solutions, and implement effective strategies for refinement.

These challenges pushed us to collaborate and think innovatively, fostering a critical approach to solving complex engineering problems. We believe that this internship offers an exceptional environment for honing problem-solving skills, which are indispensable in the engineering field.

- The mentorship we received from experienced professionals at AMSMI was instrumental in shaping our learning experience. We were guided through tasks with constructive feedback, allowing us to refine our technical skills and build confidence. The collaborative work culture encouraged us to ask questions and seek expert guidance, ensuring that we learned not just through doing, but also through mentoring and knowledge sharing.

And in order to maximize your time in the company we recommend you to do:

➤ **Focus on Real-World Problems:**

- Future interns should actively seek out real-world problems that align with the company's needs. By identifying inefficiencies or areas for improvement within the organization, you can work on projects that have a tangible impact, just as we did with the oil transfer system.

➤ **Be Proactive and Take Ownership:**

- Don't wait for tasks to be assigned take ownership of your work and propose solutions or projects. Being proactive will help you stand out and demonstrate your capability to contribute meaningful improvements to the company.

➤ **Ask Questions and Seek Guidance:**

- Always be willing to ask questions. If you don't understand something, don't hesitate to seek guidance from your supervisors or colleagues. They can offer valuable insights and advice that will help you refine your work and deepen your understanding of the industry.

➤ **Apply What You've Learned in Class:**

- Leverage the theoretical knowledge you've gained in your studies and apply it to practical situations. The internship is an opportunity to connect classroom learning with real-world applications. For example, we used our knowledge of fluid mechanics and material science to design an efficient oil transfer system.

➤ **Embrace Feedback and Learn from It:**

- Regularly seek feedback from your supervisors and peers, and be open to constructive criticism. Learning from feedback is crucial to improving your work and growing professionally.

➤ **Collaborate and Build Networks:**

- Take the time to collaborate with different teams and build relationships. Networking is an important aspect of any professional experience, and the connections you make during your internship can be valuable for your career. Engaging with people from various departments will also give you a broader understanding of how the company functions.

➤ **Be Ready to Adapt:**

- The working environment can be fast-paced, and priorities may shift. Being adaptable and open to changes in project direction or company needs is key to thriving in an industrial setting. We had to adjust our approach several times based on feedback and facility constraints, and this flexibility was crucial for the project's success.

➤ **Document Your Progress:**

- Keep detailed notes on the projects you work on, including the challenges you face and how you overcome them. This will help you when writing reports or preparing presentations, and it will be useful in showcasing your accomplishments to future employers.

➤ **Focus on Safety and Efficiency:**

- Always prioritize safety and efficiency in your projects. Understanding the company's commitment to workplace safety, especially in industrial environments, will guide you to develop solutions that not only improve processes but also adhere to safety regulations, as we did with our spill-proof oil transfer system.

➤ **Reflect on Your Experience:**

- At the end of your internship, take time to reflect on what you've learned and how it has shaped your career aspirations. Use this reflection to set clear goals for your future, whether you choose to stay in a similar industry or explore other fields.

In conclusion, we believe this internship offers an excellent opportunity for any aspiring engineer to gain hands-on experience, improve problem-solving skills, and grow professionally in a supportive learning environment. We encourage future interns to take full advantage of the mentorship and diverse projects offered, as these experiences will greatly enhance their engineering capabilities and prepare them for a successful career in mechanical or industrial engineering.

4.2.2 Recommendation for the Company:

Based on our internship experience and reflection, we propose the following recommendations to enhance the internship program at Addis Machine and Spare Parts Manufacturing Industry (AMSMI):

1. Structured Training Programs

A key recommendation is for Addis Machine and Spare Parts Manufacturing Industry (AMSMI) to establish well-structured training programs that cater to both technical and soft skills. While technical training on operating machinery like lathe and milling machines is essential, a broader

approach is required to truly prepare interns for the multifaceted demands of the engineering and manufacturing sectors.

- **Technical Skills:** We suggest that training modules include advanced instruction in machine operation, maintenance techniques, and troubleshooting common mechanical issues. This would ensure that interns are not only familiar with the basic workings of machinery but can also handle more complex tasks and malfunctions independently.
- **Soft Skills Development:** Equally important are soft skills such as communication, teamwork, and problem-solving. Manufacturing environments often require cross-departmental collaboration, and interns who are skilled in effective communication and teamwork can contribute more productively. Including training on interpersonal skills, conflict resolution, and time management will also help interns thrive in high-pressure settings and enhance the overall work environment.

A structured training program would equip interns with a balanced skill set, fostering their ability to both perform technical tasks and work effectively within a team-based industrial setting.

2. Mentorship System

The creation of a mentorship system is another critical component for enriching the internship experience. We recommend that AMSMI formalize a system where each intern is paired with an experienced professional, such as a senior engineer or technician, for the duration of the internship.

- **Guidance and Feedback:** The mentor can provide tailored guidance on day-to-day tasks, help the intern navigate challenges, and offer feedback on their progress. This personalized attention accelerates learning, enabling the intern to refine technical skills while also understanding the broader context of their role within the industry.
- **Industry Insight:** Mentors can share invaluable insights into industry standards, best practices, and emerging trends, giving interns a deeper understanding of the field. This knowledge is not always accessible through academic learning alone and can provide a competitive edge as they advance in their careers.

- **Networking Opportunities:** Establishing relationships with seasoned professionals helps interns build a professional network, which can be beneficial when seeking employment or career growth in the future.

3. Formal Feedback Mechanism:

A formal feedback mechanism is vital for continuous improvement of the internship program. We recommend that AMSMI implement a system that allows interns to provide structured feedback at different stages of their internship—such as midway and upon completion.

- **Anonymous Feedback Surveys:** Anonymous surveys can encourage interns to share honest insights about their experience, the challenges they faced, and what improvements they believe could be made. These surveys can assess various aspects of the program, including the quality of training, mentorship, workplace culture, and safety protocols.
- **Exit Interviews:** In addition to surveys, conducting one-on-one exit interviews with interns can provide more detailed feedback. These interviews can offer a deeper understanding of their learning experiences and identify areas where AMSMI can further improve its internship structure.

By implementing structured training programs, a formal mentorship system, continuous reinforcement of safety protocols, a formal feedback mechanism, and introducing cross-functional exposure, AMSMI can significantly enhance the quality and effectiveness of its internship program. These recommendations would ensure that future interns are better prepared for their roles, develop both technical and soft skills, and gain a more comprehensive understanding of the manufacturing industry. This would ultimately result in a more valuable and enriching experience for interns, while also benefiting AMSMI through improved intern performance and engagement.